A Survey of Large Language Models: Progress, Challenges and Future Directions

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Abstract

Large language models have become the dominant paradigm in natural language processing and are rapidly spreading to vision, speech, and scientific discovery. This survey organizes the field along four axes: architecture and pre-training, adaptation and alignment, reasoning and evaluation, and deployment. We review the progression from the Transformer to modern decoder-only models with hundreds of billions of parameters, and we trace the parallel development of vision transformers, parameter-efficient adaptation, and retrieval-augmented generation. We then discuss the open challenges of factual reliability, long-context reasoning, efficiency, safety, and evaluation, and we outline five future research directions. Our goal is to provide a coherent map of the field for researchers entering it and for practitioners choosing architectures and adaptation strategies.

Keywords: large language models, survey, transformer, adaptation, reasoning, evaluation

1 Introduction

Large language models have changed the practice of natural language processing. A single pre-trained model can now be adapted to translation, summarization, question answering, code generation, and dialogue with little or no task-specific training. This success has been driven by three factors: the Transformer architecture, large-scale pre-training on web-scale corpora, and scaling laws that predict performance from compute and model size.

The field has grown quickly, and the literature is now fragmented across architecture papers, adaptation methods, evaluation benchmarks, and deployment reports. This survey provides a unified view. Section 2 reviews the foundations, Section 3 compares model families, Section 4 covers adaptation, Section 5 discusses reasoning and evaluation, Section 6 surveys applications, Section 7 lists challenges, and Section 8 lays out future directions.

Our scope is deliberately broad. We include encoder-only models such as BERT, decoder-only models such as the GPT series, vision transformers, parameter-efficient methods such as LoRA, and retrieval-augmented models such as RAG. We emphasize ideas that have proven durable rather than reporting every benchmark number, and we point readers to the original papers for experimental details.

2 Background and Foundations

The modern era of language modeling begins with the Transformer, which replaced recurrence with self-attention and made training on long sequences parallelizable. Its encoder-decoder form was quickly adapted into encoder-only and decoder-only variants, and these three families remain the basis of most models in use today.

2.1 The Transformer Architecture

The Transformer stacks multi-head self-attention and position-wise feed-forward layers with residual connections and layer normalization. Its key property is that every position can attend to every other position in a constant number of operations, which removes the sequential bottleneck of recurrent networks. The original model used six encoder and six decoder layers, eight attention heads, and a hidden size of 512, and it established state of the art on WMT translation.

Two design choices proved especially influential. First, the decoder-only variant with causal masking supports efficient autoregressive generation and scales cleanly. Second, the same block can be applied to images by treating image patches as tokens, which led to the vision transformer. We return to both in Section 3.

2.2 Pre-training Objectives

Encoder-only models such as BERT are trained with masked language modeling, in which a fraction of input tokens is replaced by a mask and the model

predicts the original tokens. BERT also uses next-sentence prediction, and it established that bidirectional pre-training transfers well to downstream classification and span extraction. Decoder-only models instead predict the next token on a left-to-right prefix, which yields a single objective that supports both zero-shot prompting and generation.

The choice of objective shapes what the model is good at. Masked modeling produces strong representations for understanding tasks but is awkward for generation, while causal modeling produces a general-purpose generator but weaker sentence representations. Recent work attempts to combine the two, for example by using span corruption or by training with multiple objectives.

2.3 Scaling Laws

Empirical scaling laws describe how loss decreases as a power law in model size, dataset size, and compute. These laws motivated the jump from hundreds of millions to hundreds of billions of parameters, including GPT-3 with 175 billion parameters and PaLM with 540 billion. They also motivate compute-optimal training, which balances parameters and tokens rather than maximizing model size alone.

Scaling laws do not by themselves explain emergent abilities, such as multi-step reasoning, that appear at particular scales. Whether emergence is a genuine discontinuity or an artifact of benchmark metrics is an open question, and it is one of the motivations for the evaluation discussion in Section 5.

A third axis is inference-time computation. Rather than increasing the number of parameters, a system can spend more compute while answering a single query by sampling several candidate chains, searching over intermediate steps, or voting across samples. Early evidence suggests that test-time compute can substitute for some of the gains normally obtained by scaling the model, particularly on reasoning tasks that reward exploration. This complicates the classical scaling-law picture, because the effective compute budget of a deployed system now includes both training and inference, and the optimal balance between the two is task dependent.

3 Model Architectures

We group models into encoder-only, decoder-only, and vision transformer families. The grouping is useful because each family has a characteristic training objective, adaptation recipe, and set of strengths.

3.1 Encoder-Only Models

BERT is the canonical encoder-only model. The base model has 12 layers, a hidden size of 768, and 12 attention heads, for 110 million parameters, while the large model has 24 layers, a hidden size of 1024, and 16 heads, for 340 million parameters. It was pre-trained on BooksCorpus and English Wikipedia with masked language modeling at a 15 percent mask rate, and it was trained on 64 TPU chips for four days.

Encoder-only models remain the best choice for tasks that require a single representation of the input, such as classification, retrieval, and span extraction. Their main limitation is that they are not directly generative, so they must be paired with a decoder or used as a component in a larger system, as in retrieval-augmented generation.

BERT is trained with two objectives: masked language modeling, which predicts randomly masked tokens from both sides of the context, and next sentence prediction, which classifies whether two segments are consecutive. Later work showed that next sentence prediction contributes little and that more data and longer training matter more than architectural changes. RoBERTa removes next sentence prediction, trains on ten times more data, and uses dynamic masking; it matches or exceeds BERT on most benchmarks with the same architecture. This line of work established a pattern that continues today: once a strong architecture exists, gains come mainly from data, scale, and training procedure.

3.2 Decoder-Only Models

The GPT series established the decoder-only paradigm. GPT-1 used 117 million parameters, 12 layers, a hidden size of 768, and 12 heads, and it was pre-trained on BooksCorpus with 7000 books and then fine-tuned on downstream tasks. GPT-2 scaled to 1.5 billion parameters, 48 layers, a hidden size of 1600, 24 heads, and a context length of 1024, and it was trained on WebText, a corpus of 8 million documents totaling 40GB.

GPT-3 scaled to 175 billion parameters, 96 layers, a hidden size of 12288, 96 heads, and a context length of 2048. It was trained on 10000 NVIDIA V100 GPUs and introduced in-context few-shot learning, in which the model performs a task from examples in the prompt without any gradient updates. This shift from fine-tuning to prompting reorganized the field around a single general-purpose model.

3.3 Vision Transformers

The vision transformer applies the Transformer block to images by splitting an image into 16x16 patches, embedding each patch, and treating the resulting sequence as tokens. The largest variant, ViT-H/14, has 632 million parameters and 32 Transformer layers, and it reaches 88.55 percent top-1 accuracy on ImageNet when pre-trained on JFT-300M. Training used 32 TPUv3 cores.

Vision transformers showed that the architectural advantage of attention is not specific to text. They also illustrated the importance of data scale: on small datasets, convolutional networks such as ResNet remain competitive, and the transformer advantage appears only with large-scale pre-training.

4 Adaptation and Alignment

Pre-training is only the first stage. Adapting a model to a task, a domain, or a set of human preferences is now a research area in its own right, because full fine-tuning becomes expensive as models grow.

4.1 Fine-Tuning and Instruction Tuning

Full fine-tuning updates all parameters on a task-specific dataset and remains a strong baseline. Instruction tuning instead trains on a mixture of tasks phrased as instructions, which improves zero-shot generalization to unseen instructions. Reinforcement learning from human feedback adds a reward model trained on human comparisons and is widely used to align model outputs with user preferences.

4.2 Parameter-Efficient Adaptation

LoRA freezes the pre-trained weights and learns a low-rank update for each adapted matrix. On GPT-3 175B it trains only 4.7 million parameters with rank r = 8 and scaling alpha = 16, reduces GPU memory by 3 times, and adds no inference latency because the update can be merged into the weights. Adapter layers and prefix tuning are related methods that insert small trainable modules or prepend trainable vectors, but both add inference cost.

Parameter-efficient methods are now the default for adapting large models to many tasks, because they make it practical to store and serve thousands of task-specific variants from a single base model.

From a systems perspective, parameter-efficient adaptation changes the economics of serving. A single base model can remain resident in memory while many small adapters are swapped per request, which reduces the memory footprint relative to keeping one full model per task. Adapters can also be versioned and rolled back independently of the base weights, which simplifies deployment. The main open questions are how to compose several adapters without interference and whether low-rank updates remain sufficient as base models continue to grow.

4.3 Retrieval Augmentation

Retrieval-augmented generation pairs a parametric generator with a non-parametric index. RAG retrieves the top 5 documents from a dense index of 21 million Wikipedia passages and conditions a FiD-style generator on them, reaching 44.5 exact match on Natural Questions with a DPR retriever. Because the knowledge lives in the index, it can be updated without retraining the model, and the retrieved passages provide provenance for the answer.

Retrieval also complements long-context models: instead of forcing the model to attend over hundreds of thousands of tokens, a retriever can select the relevant passages first. The two approaches are likely to be combined rather than to replace each other.

5 Reasoning and Evaluation

Reasoning and evaluation are tightly coupled: as models improve on standard benchmarks, the benchmarks themselves become saturated, and new evaluations are needed to measure progress.

5.1 Prompting and Chain-of-Thought

Chain-of-thought prompting asks the model to produce intermediate reasoning steps before the final answer. With PaLM 540B, it raises the GSM8K solve rate from 17.9 percent to 56.9 percent, and it produces large gains on commonsense and symbolic reasoning tasks. The gains are an emergent property of scale: they are small for models below roughly 100 billion parameters.

Chain-of-thought prompting is not a guarantee of faithful reasoning. Models can produce correct answers from incorrect chains, and they can produce plausible chains that lead to wrong answers. Verifying reasoning traces is therefore an active research problem.

Prompting has also become a design interface for applications. A fixed prompt can be evaluated and versioned like code, and small changes in wording or examples can shift accuracy by several points. This sensitivity makes prompt engineering labor intensive and motivates automatic prompt optimization, which searches over instructions and demonstrations using a validation set. Even so, prompts are brittle across model versions, so production systems usually combine prompting with retrieval and with lightweight fine-tuning of the base model.

5.2 Benchmarks and Their Limits

Benchmarks such as MMLU, GSM8K, HumanEval, and HELM measure different capabilities, and no single benchmark captures general competence. Contamination between pre-training data and test sets is a persistent concern, and leaderboard overfitting can make progress look faster than it is. We recommend reporting evaluation details, including prompt format, decoding settings, and contamination checks.

6 Applications

Large language models are now used in search, customer support, software engineering, education, and scientific research. In search and question answering, retrieval-augmented models provide answers with citations. In software engineering, models generate, explain, and test code. In science, they assist with literature review, data extraction, and hypothesis generation.

Deployment patterns differ by application. High-stakes domains such as medicine and law require human review and strong provenance, while low-stakes domains can tolerate more automation. The common requirement is evaluation: every deployment needs a task-specific test set and a monitoring plan for distribution shift.

Multimodal applications are the next frontier. Models that accept images, audio, and text are being deployed in accessibility tools, medical imaging, and robotics. These systems raise new evaluation questions, because correctness may depend on the interaction between modalities, and they require data pipelines that align modalities at scale. Early results suggest that the same transformer backbone can serve all of them, which is one reason the architecture has proved so durable.

7 Challenges

The first challenge is factual reliability. Models can hallucinate fluent but incorrect content, and retrieval reduces but does not eliminate this problem. The second challenge is long-context reasoning: attention cost grows quadratically with sequence length, and models do not reliably use information in the middle of long contexts.

The third challenge is efficiency. Training and serving hundred-billion-parameter models require substantial compute, and parameter-efficient adaptation only partly addresses the cost. The fourth challenge is safety and governance: models can produce harmful content, leak private data, or be misused, and the norms for auditing and disclosure are still being developed.

The fifth challenge is evaluation. Static benchmarks saturate, contamination is hard to rule out, and human evaluation is expensive. Progress on all five challenges requires better measurement as much as better models.

A sixth challenge is data governance. Pre-training corpora are large, poorly documented, and often contain personal or copyrighted material, which creates legal and ethical exposure for researchers and deployers alike. Data provenance, opt-out mechanisms, and reproducible documentation of filtering decisions are therefore becoming part of model development rather than an afterthought. Addressing this challenge requires collaboration between technical teams, legal experts, and the communities whose data is used.

8 Future Work

We identify five directions that we expect to shape the next phase of research. First, efficient long-context architectures. Attention alternatives, state-space models, and retrieval hybrids should be compared under a common protocol that measures both quality and cost at context lengths beyond one million tokens.

Second, faithful reasoning and verification. Chain-of-thought traces are useful but not guaranteed to reflect the model's computation. Future work should develop verifiers that check intermediate steps, and should study when generated explanations can be trusted.

Third, retrieval and tool use. Retrieval-augmented generation has shown that external memory improves factuality, but current systems retrieve once and generate once. Future systems should interleave retrieval, tool calls, and generation, and should learn when to retrieve and when to rely on parameters.

Fourth, multimodal grounding. Vision transformers and multimodal models have made rapid progress, but grounding language in perception, action, and structured data remains difficult. Benchmarks that require spatial, temporal, and causal reasoning across modalities are needed.

Fifth, safety, evaluation, and governance. As models are deployed in high-stakes settings, we need auditable evaluations, provenance for generated content, and mechanisms for redressing harm. Technical work on watermarking, uncertainty estimation, and interpretability should be paired with policy work on disclosure and accountability.

9 Conclusion

Large language models have advanced rapidly because architecture, data, and scale improved together. The Transformer provided a scalable block, pre-training provided a general objective, and scaling provided predictable gains. Adaptation methods, retrieval, and prompting then turned a single pre-trained model into a platform for many tasks.

The remaining challenges are substantial. Factual reliability, long-context reasoning, efficiency, safety, and evaluation all require sustained work. We hope that this survey provides a useful map of the field and a clear statement of the open problems that matter most.

$$\text{Loss} = -\sum_t \log p(x_t \mid x_{<t}) \tag{1}$$

$$h = W_0 x + \frac{\alpha}{r} B A x \tag{2}$$

$$p(y \mid x) = \sum_{z \in \text{top-}k} p(z \mid x)\, p(y \mid x, z) \tag{3}$$

Table 1: Representative model families and their scale. Parameter counts are for the largest reported variant.

Family	Model	Params	Context
Encoder-only	BERT-large	340M	512
Decoder-only	GPT-2	1.5B	1024
Decoder-only	GPT-3	175B	2048
Vision	ViT-H/14	632M	196 patches
Adaptation	LoRA on GPT-3	4.7M trainable	2048

Table 2: Selected results discussed in this survey.

Method	Task	Result
BERT-large	GLUE average	80.5
GPT-3 175B	LAMBADA accuracy	86.4
ViT-H/14	ImageNet top-1	88.55
ResNet-152	ImageNet top-5	3.57
RAG	Natural Questions EM	44.5
PaLM 540B + CoT	GSM8K solve rate	56.9

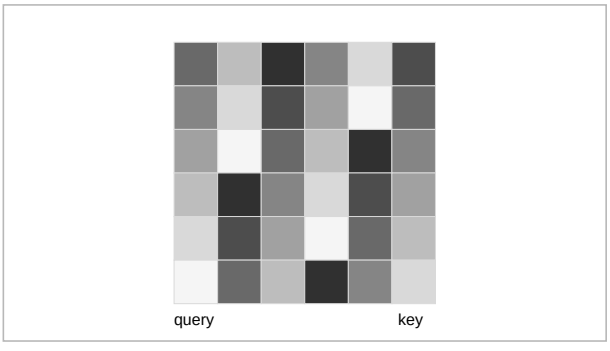


Figure 1: Timeline of model families covered in this survey, from the Transformer in 2017 to retrieval-augmented and parameter-efficient models in 2024.

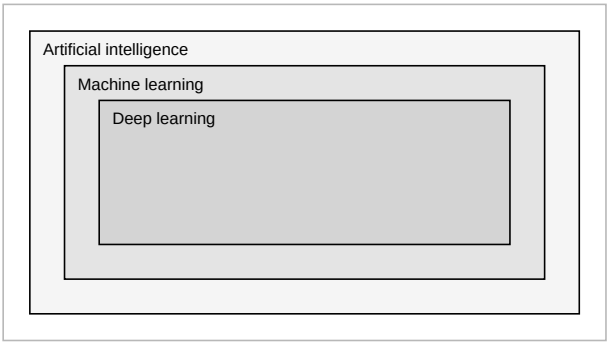


Figure 2: Taxonomy of adaptation methods: full fine-tuning, instruction tuning, parameter-efficient methods, and retrieval augmentation.

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A Appendix: Method Notes

This appendix records how papers were selected and how numbers were checked. We searched for architecture, adaptation, reasoning, and evaluation papers published between 2012 and 2024, and we retained papers that introduced a widely used method or that reported a result cited by later work. We gave priority to papers with public code and to papers whose results have been reproduced.

All parameter counts and benchmark numbers in the survey are taken from the original papers and are reported for the largest variant unless stated otherwise.

Where a paper reports multiple settings, we use the setting that the authors describe as their main result. We did not re-run any experiments, so the numbers should be read as reported values rather than as independent replications.

The five future directions in Section 8 were chosen by clustering the limitations discussed in the papers we reviewed. They are not intended to be exhaustive. We expect the relative importance of the directions to change as models, benchmarks, and deployment practices evolve.

B Appendix: Glossary

Fine-tuning: updating all model parameters on a task-specific dataset. Instruction tuning: fine-tuning on a mixture of tasks phrased as instructions. Parameter-efficient adaptation: adapting a model while training only a small number of additional parameters. Retrieval augmentation: conditioning generation on documents retrieved from an external index. Chain-of-thought prompting: few-shot prompting in which exemplars include intermediate reasoning steps.